

FlexiTech Electronics: Supply Chain Strategy Analysis

Group 9

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Deliverable 1: Root-Cause Diagnosis- Noelia Torres Lopez

Deliverable 2: Postponement Design Recommendation- Zain Amanullah

Deliverable 3: Quantified Impact and Economics- Yangqing Jin

Deliverable 4: Risks and Implementation Roadmap- Eshika Shrivastava

Deliverable 1

FlexiTech Electronics is currently running out of its most popular laptop configurations while simultaneously sitting on excess inventory of configurations that aren't selling. When analyzing the situation, it's clear FlexiTech has three root causes contributing to this issue.

The first root cause is the extensive variety of products. According to exhibit 3, the high velocity core segment accounts for 65% of unit volume while representing only 20% of total SKUs, meaning that only 20% of their product configurations are driving the majority of sales. In addition, forecast error increases substantially for lower-volume products, reaching as high as 60–90% for long-tail SKUs. Due to this, FTE has greater difficulty accurately forecasting demand, which in turn leads to excess inventory of slow-moving configurations and stockouts of popular products. Exhibit 7 further reinforces this, stating that 23% of retail inventory has been sitting for over 180 days, which is nearly double the 15% target. That's a direct sign that FTE is building configurations that customers simply do not want to buy.

The second root cause is that product configurations are locked in too early. Exhibit 2 shows that RAM variants and SSD variants are completed by the contract manufacturer before the products are shipped to the United States. In addition, software images and labeling are completed upstream, meaning these decisions are made before products reach the U.S. market (The Supply Chain Consulting Group, 2026). As a result, FTE has to determine the exact configuration of each product, before actual demand is known. Due to forecasts being unpredictable, the company is producing too many low-demand configurations while not producing enough high-demand configurations. Because these components and specifications

have already been committed, FTE has limited flexibility to adjust inventory once products have been manufactured and shipped. This problem gets even worse when you factor in the timeline. Based on Exhibit 4, the end-to-end supply chain process from component procurement to US distribution center receipt, takes anywhere from 64 to 106 days. By the time FTE realizes they built the wrong mix, the product is already in transit.

The third root cause of FTE's current circumstances is the long and inflexible lead times. Based on Exhibit 4, the end-to-end supply chain process spans multiple stages including component procurement, factory build, ocean transit, customs, and inland transport, each adding time and variability that FTE has no control over. Even under normal conditions, the lead time for a product to travel from the factory to a US distribution center can take up to 6 to 8 weeks. During peak seasons, delays from port congestion and inland transportation can push that timeline even further, making it more difficult for FTE to meet customer expectations and delivery requirements. Exhibit 7 further highlights the impact of these long lead times, with retail fill rates at only 88% and OTIF performance at 82%, both falling below FTE's targets of 95%.

In conclusion, FTE's inventory challenges stem from excessive product variety, product configurations being committed too early, and long, inflexible supply chain lead times. Together, these factors increase forecast error, reduce supply chain flexibility, and prevent FTE from responding to changing customer demand. For FTE to move forward, the company must reduce configuration complexity and prioritize availability of its most popular configurations, push final product decisions closer to when demand is actually known, and build more flexibility into its supply chain. At the same time, any improvements must be made without sacrificing service levels across retail, DTC, and institutional channels, as FTE cannot afford to lose any more

ground with its retail partners. FTE also needs to be mindful of costs, as spending more money on expedited shipping or emergency orders is not a sustainable solution to a problem that keeps repeating itself. Finally, FTE must reduce the amount of cash tied up in slow moving inventory that has been sitting in warehouses for over 180 days, as that working capital could be better used elsewhere in the business.

Deliverable 2

FTE experiences simultaneous stockouts and excess inventory, resulting from long lead times, excessive SKU variety, and having configurations locked in too early at the factory, leaving FTE unable to match supply to demand. As a solution FTE should adopt option B, hardware postponement at its Dallas facility, to address this issue. While option A addresses software image labeling and final packout, it does not resolve the RAM and SSD proliferation identified within exhibits 2 and 3, a primary driver of FTE's inventory mismatch.

Today production is outsourced to contract manufacturers in Vietnam and Malaysia. This is where FTE's decoupling point, meaning all configuration decisions, such as RAM and SSD variants, are committed at the factory before products begin their journey to U.S. distribution centers. In addition, there is a lead time of 6-8 weeks to transport products to U.S. distribution centers. Specifically as outlined in exhibit 4 the lead time for key components like RAM and SSD is 21-35 days, while the end to end lead time from component procurement to distribution center receipt ranged from 64 to 106 days under normal conditions. Once finished goods have arrived and are ready to go demand signals have already shifted, which means that FTE is already committed to configurations made at the factory several weeks prior. Consequently, supply and demand matching problems arise, however, option B can help by moving the

decoupling point downstream to the Dallas facility. Within this option FTE imports a smaller set of common core units and postpones RAM insertion, SSD installation, software imaging, labeling, and final packout at Dallas in response to actual orders. This change allows FTE to delay high variance configuration decisions later downstream to help align closer with real time demand, directly addressing the root cause of inventory mismatches.

Option B is the right choice for FTE's channel mix and product architecture for several reasons. Exhibit 3 shows SKU velocity and planning complexity. Within this visual it shows that the top 20% of SKUs from segment A account for 65% of unit volume, with forecast errors of 20-30%. The next 35% of SKUs from segment B account for 25% of volume with MAPE of 35-50%. Segments A and B represent 90% of volume and are managed as finished goods. Option B allows FTE to hold common core inventory for these segments and differentiate at the facilities in Dallas instead of around 2 months in advance. Exhibit 2 also supports this choice by confirming that RAM and SSD are responsible for SKU proliferation and that they are postponable at the Dallas facility. The chassis and motherboard are not postponable and remain upstream. On the other hand, option A only postpones software imaging and labeling, which address specific compliance needs. However, it leaves the RAM and SSD configurations still committed at the factory. Availability issues that occur since 16GB RAM and 1TB SSD are identified as high velocity combinations where stockouts occur cannot be resolved by option A that triggered FTE's correction plans from a major retail partner.

For option B to be a successful strategy for FTE there must be several process capabilities put into place. First off, an ESD safe workstation is needed for RAM and SSD insertion. This is because modern chips operate at ultra low voltages making them susceptible to

electrical overstress resulting in physical harm or product failures over time (IEEE Innovation at Work, 2025). Improper handling of components or failures can damage customer satisfaction and increase costs, which is why ESD safe areas are so important. Second, a kitting and staging operation is needed to ensure that RAM and SSD combinations are matched to each other. By utilizing product kitting the right components will be matched with one another and it reduces the risk of missing parts (Schlecht, 2024). Also, it is important to have a post-assembly functional test capability. While the Dallas facility performs incoming inspections and testing, it needs to be expanded to validate newly inserted components. Fourth, a software imaging station must be in place to apply the right image per channel with audit logs and controls for institutional buyers. Finally, there must be labeling capabilities to track assets and have proper documentation.

The bottleneck step is likely to be the post-assembly functional testing. Functional testing requires trained technicians, dedicated equipment, and fixed cycle times per unit. Other process capabilities such as imaging and kitting can be automated to a higher degree, however, functional testing is a more manual and critical restraint. Particularly given that FTE's demand is highly seasonal, if the test capacity is lower than peak volume, Dallas facilities can become a bottleneck that causes a new problem while solving an old one. In order to avoid this, testing should be a high priority in the build of the facility and it should be capable of handling segment A and B volume, while taking seasonal demand into account.

Overall, option B is a better postponement choice because it addresses the configuration level root inventory problem, addresses the modular architecture and postponable elements through the proper capabilities, and it aligns with the channel mix that causes the most risk.

Moving the decoupling point from Malaysia and Vietnam to Dallas brings FTE's decisions closer to demand providing flexibility that was previously not there.

Deliverable 3

In this case study, I will assume that FlexiTech Electronics chooses to implement a hardware deferral strategy at its Dallas facility. Under this strategy, FlexiTech will first import standard models of the BizBook 14 and then complete the final configuration in Dallas. The Dallas facility will handle the installation of memory and solid-state drives, software imaging, label application, and final packaging. This approach is highly suitable because memory and SSDs are classified as components that can be deferred. However, decisions regarding the CPU and Wi-Fi cannot be deferred, as they are tied to supplier allocation quotas and must be determined in advance.

The core economic rationale is that FlexiTech should not stockpile excessive finished SKUs until actual demand is clear. Instead, the company can maintain a small number of flexible, generic core models and complete assembly at a later stage based on customer or channel demand. This will help FTE better align with actual demand, reduce slow-moving inventory, and ensure a sufficient supply of top-tier configurations during promotional periods. The following analysis will evaluate the potential impact of this approach on inventory turnover, service performance, and total costs.

First, I will discuss inventory turns. FTE currently has about 2.5 inventory turns per year, which is much lower than the 5-6 turns reported by some direct competitors. This shows that the problem is not only that FTE has too much inventory, but also that the company has the wrong type of inventory. Exhibit 3 shows that only 20% of SKUs create 65% of unit volume. At the

same time, the long-tail segment makes up 45% of SKUs but only 10% of total volume. This long-tail segment also has a very high forecast error, with a 60–90% MAPE, so it is risky for FTE to keep these products as finished goods. Reducing aged inventory is also very important for FTE. Exhibit 8 shows that 170,000 units are more than 180 days old, representing 34% of the total units. These older units can usually be sold for only 40–70% of their value if liquidated. If FTE can reduce aged inventory to around 15–20%, it would be much closer to the retailer's requirement of keeping aged inventory below 15%. It would also help the company reduce markdowns and inventory write-downs.

Second, is the service performance. The current service gaps also pose a serious problem for FTE. In the retail channel, the order fulfillment rate is only 88%, whereas the target is 96%. The on-time-in-full (OTIF) rate in the retail channel is also low, at just 82%, far below the 95% target. The institutional channel performs slightly better in order fulfillment, reaching 93%, but its OTIF rate remains at 86%. These gaps are critical because a major retailer has included FTE in its remediation plan. This retailer has also tied future shelf space and promotional opportunities to FTE's ability to improve product availability and on-time delivery rates.

My option helps improve service levels because FTE does not need to prematurely determine the combination of each RAM, SSD, and software image before actual demand is clear. Instead, FTE can maintain generic core units and finalize configurations in Dallas based on actual customer or channel demand. The greatest improvement in service levels should come from support for high-turnover retail configurations and from reducing direct order cancellation rates for consumer-facing orders. Improving supply capabilities in this segment will significantly improve

overall service performance. Institutional customers will also benefit, as the Dallas team gains better control over software images, labeling, serialization, and document management.

Last is the total cost. The financial impact of my option mainly comes from reducing three cost categories: markdowns and price protection, expedited freight, and inventory write-downs. According to Exhibit 8, these costs are expected to reach \$52 million in 2026. This includes \$26 million for markdowns and price protection, \$14 million for expedited freight and premium logistics, and \$12 million for inventory write-downs. These numbers show why FTE is under strong financial pressure. Even though the company is projected to have \$640 million in net sales in 2026, its operating income is still expected to be negative \$13 million. This means that the inventory and service problems are not only operational issues, but also major financial problems for the company.

Finally, there is the discussion of sensitivity. The recommendation depends most on three main assumptions. First, Dallas needs reliable conversion yields and cycle times. If RAM/SSD insertion, software imaging, and testing become bottlenecks, then the service improvement may be much smaller. Second, reducing aged inventory is a major contributor to the financial benefit. If FTE continues to buy the same finished-goods mix and only adds extra Dallas work, the cost savings will not be very strong. Third, the recommendation assumes that RAM/SSD and software image variety are still the main reasons for the mismatch between supply and demand. If CPU or Wi-Fi allocation becomes the biggest constraint, Dallas postponement would help less because those parts cannot be postponed in the same way.

Deliverable 4

In order to keep up with lead times, FlexiTech Electronics will make the decision to implement postponement with Dallas. This would mean that the company would customize the product in Dallas when there was a closer demand for it. This can help improve lead times and reduce many of the issues we were facing such as stockouts and excess inventory. However, this solution can come with many risks. Currently, Dallas only handles things such as incoming inspection, functional testing, and limited warranty triage. By implementing postponement with Dallas, they will take on more opportunities such as software imaging, labeling, and advanced design such as RAM/SSD insertion. Unfortunately, this expansion can come with many risks such as poor quality of products, delays in testing, issues with training workers on advanced design, as well as a possibility of the Dallas locations being overwhelmed with orders and design.

We can mitigate these risks by hiring experienced workers, and making sure Dallas has all the resources it needs for this transition to result in a successful postponement plan. The most likely risk is a risk in the overall quality of the product. Right now the Dallas plant doesn't have any experience with advanced design and installing RAM and SSDs since this used to be outsourced to manufacturers in Vietnam and Malaysia. To mitigate this risk, FlexiTech can set up extensive training for their employees making sure they are all equipped to do this. Additionally, for the first 2 weeks of this transition, they can hire an expert in this process to supervise employees that go through this training to make sure they are not making any mistakes when installing this technology. Another possible risk is delays in testing due to having a lack of workers or just not having the full capacity to test each unit while keeping up with demand. There is a possibility that testing can be a bottleneck and slow everything else down. One way

we can reduce this risk is by making sure the Dallas location is big enough to do testing and handle this expansion. We can evaluate whether we need to expand the Dallas location since they will be doing much more than just inspection and testing. Another risk that we can face is issues with training the workers, which could lead to installation errors. Having a structured training process can help mitigate this issue because it can ensure the least amount of errors are being made. Employees can also receive certifications for knowing how to install software to ensure that everyone installing software is trained efficiently. Finally the last risk is Dallas reaching its capacity limits. Having the additional responsibility of customization can lead to a high order volume causing many orders to not be fulfilled. This is a very high risk that not only can result in the issue being unsolved, but it can make the issue worse for FlexiTech. Some ways we can mitigate this is by expanding the Dallas location and to phase out postponement so we can slowly receive order rather than receive demand all at once.

For our proposed roadmap, we would like to split this expansion into 3 different phases: our planning and training phase (0-90 days), our pilot phase (90-180 days), and then our full rollout phase(180 to 360 days). For phase one we want to focus on planning for the expansion, so this would include evaluating our current space and deciding if we need to expand, making any changes to the space and implementing an extensive training program to make sure all of our employees are trained to handle advanced design. For phase 2, our pilot phase, our main goal is to start the postponement process and only handle a set amount of orders. In this phase we want to see if our employees are able to handle the demand we face and increase the number of orders we take based on that. We also want to see if this postponement process is working, if our employees need any more training, or if we need to change anything about our system to prevent

any errors that can happen in the next phase (rollout). In phase 3 we want to fully expand all of the design orders that would be outsourced to happen in our Dallas location. This would be a full rollout as well as analyzing if this is a sustainable system for us in the long term future. In this rollout phase we will be handling around the same quantity of orders that we would be sending out to our suppliers overseas. By implementing a phased postponement we can get ahead of any risks that we might encounter and improve our lead times by a significant number.

We will have some Key Performance Indicators (KPI's) to make sure we are on the right track with our postponement plan. Our primary risk is poor quality and a KPI that we can set to monitor that is evaluating the quality pass rate of every unit. This number will tell us how many products were made correctly and help us make sure that we don't send out faulty units. Another KPI we can look at is testing dates and analyzing how long they take to see if there are any delays. In order to ensure all of our workers are trained, we can look at what percentage of the current employees are trained and how long their certification took them. Some KPIs we can look at when it comes to order fulfilment is looking at the lead times for each order and seeing if there are any delays in shipping out our orders. This can tell us if our new plan is actually working and if our lead times are decreasing.

In conclusion, by implementing postponement in Dallas, FlexiTech can solve some of their performance issues. Although there are risks to this postponement plan, such as poor quality of products, delays in testing, issues with training workers on advanced design, as well as a possibility of the Dallas locations being overwhelmed with orders and design, they can be mitigated by phasing out the postponement plan making sure that this transition is seamless. By

analyzing the KPIs listed above we can monitor whether this plan is actually working and adjust the plan accordingly.

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